

Poisson Distribution

$X \sim \text{Poisson}(\lambda)$, Used to model how many times a certain event occurs in a specific time interval or in a specific length or area.

- Events which occur in time intervals that do not overlap are **independent**.
- Mean Occurrence or Arrivals of events in time interval $= \lambda$ should be **constant**.

Examples:

Suppose there are 3.2 Ferrari cars passing Orchard Road in every 15 mins.
Let X be the number of Ferrari cars passing Orchard Road in **one hour**.
i.e. $X \sim \text{Poisson}(3.2 \times 4)$.

Suppose on average there are 2.3 errors in 10 pages of a report.
Let Y be the number of errors found in **100 pages** of report.
i.e. $Y \sim \text{Poisson}(2.3 \times 10)$.

Pmf of Poisson distribution

The **Poisson frequency function** with parameter $\lambda (\lambda > 0)$ is

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad \text{for } k = 0, 1, 2, \dots$$

Approximation of Binomial Distribution

The Poisson distribution can be derived as the limit of a binomial distribution with $np = \lambda$, as $n \rightarrow \infty$. i.e.

$$X \sim \text{Bin}(n, p) \text{ can be } \underline{\text{approximated}} \text{ as } X \sim \text{Poisson}(np)$$

Expectation and Variance of $X \sim \text{Poisson}(\lambda)$

$$\mathbb{E}(X) = \lambda, \quad \text{Var}(X) = \lambda.$$

For approximation of $Y \sim \text{Bin}(n, p)$

$$\mathbb{E}(Y) = np \rightarrow \lambda, \quad \text{Var}(Y) = np(1 - p) \rightarrow \lambda$$

when $n \rightarrow \infty$. (Remark: This approximation is good if $n \geq 50$ and $np \leq 5$.)

Geometric Distribution

Given n independent trials. On each trial, a success occurs with probability p .

Let X be the total number of trials up to and including the first success,
 $X \sim \text{Geometric}(p)$

If $X = k$, then there must be $k - 1$ failures preceding the first success.

$$P(X = k) = (1 - p)^{k-1}p, \quad \text{for } k = 1, 2, \dots$$

Examples: Flipping a dice until it lands on '6'. i.e. $X \sim \text{Geometric}(\frac{1}{6})$
Suppose a researcher ask people if they support a certain law. The probability that a given person supports the law is 0.2. Let Y be the number of people he would need to ask until he get a supporter of the law.
 $Y \sim \text{Geometric}(0.2)$.

Expectation and Variance of $X \sim \text{Geometric}(p)$

$$\mathbb{E}(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}.$$

Hypergeometric Distribution

Suppose an urn contains N balls, of which m are white and $N - m$ are black. Let X denote the number of white balls drawn when taking n balls *without* replacement. $X \sim \text{Hypergeometric}(n, N, m)$.

$$P(X = k) = \frac{\binom{m}{k} \binom{N-m}{n-k}}{\binom{N}{n}}$$

- n can be seen as the sample.
- N can be seen as the total (i.e. # of black + # of white).
- m can be seen as # of whites.

Expectation and Variance of $X \sim \text{Hypergeometric}(n, N, m)$

$$\mathbb{E}(X) = n \cdot \frac{m}{N}, \quad \text{Var}(X) = n \cdot \frac{m}{N} \left(1 - \frac{m}{N}\right) \left(1 - \frac{n-1}{N-1}\right).$$

Expectation of sum of random variables

Let $X_1, X_2, \dots, X_n$ be random variables on the same sample space Ω , then

$$\mathbb{E} \left(\sum_{i=1}^n X_i \right) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \dots + \mathbb{E}(X_n) = \sum_{i=1}^n \mathbb{E}(X_i).$$

Note that this is particularly useful when you treat X_i as events of Indicator Random Variables.

Indicator Random Variable

I is an indicator variable for the event A if

$$I = \begin{cases} 1, & \text{if } A \text{ occurs;} \\ 0, & \text{if } A \text{ does not occur.} \end{cases}$$

The indicator variable for A is also denoted by $\mathbf{1}_A$.

$$\mathbb{E}(I) = 0 \cdot P\{I = 0\} + 1 \cdot P\{I = 1\} = P(A).$$